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A modified entropy functional for improved high order shock capturing methods

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ABSTRACT

This study proposes a new piecewise entropy functional aimed at improving entropy based troubled element detection in high order shock capturing methods under demanding near vacuum conditions. In such regimes, the classical thermodynamic entropy $\log(p/\rho^{\gamma})$ becomes singular and highly sensitive to small perturbations of the isentropic pressure, which can compromise the reliability of entropy thresholding strategies. The proposed functional is designed to reduce this stiffness while retaining the classical logarithmic entropy behavior away from the low isentropic pressure regime. Building upon the generalized entropy framework introduced by Harten, the proposed functional is analyzed in terms of compatibility and convexity. The analysis shows that it behaves as a valid convex entropy flux pair above a computable threshold in the isentropic pressure. Below this threshold, where strict convexity is lost, the functional is not interpreted as the basis for a global entropy stability proof, but rather as a regularized entropy indicator for troubled element detection. The functional is constructed to be piecewise continuously differentiable and is specifically tailored for use within modal filtering based shock capturing strategies for Discontinuous Galerkin methods. The only modification to the baseline filtering framework is the replacement of the original thermodynamic entropy used in the detection criterion with the proposed piecewise functional. Numerical experiments demonstrate that the modified entropy improves the reliability of troubled element detection in challenging near vacuum and high Mach number configurations, while preserving the robustness of the underlying shock capturing procedure.

1. Introduction

Entropy plays an important role in the context of computational fluid dynamics (CFD). Early efforts, rooted in the generalized entropy framework of Harten [1], defined a numerical entropy functional along with an associated entropy flux, requiring convexity of the functional and compatibility with the underlying conservation laws [1,2]. Building on these ideas, Tadmor introduced entropy conservative and entropy stable schemes by deriving discrete fluxes that exactly satisfy a discrete analogue of the continuous entropy balance, thereby preventing non physical oscillations across discontinuities [3,4]. This approach has also been extended to essentially non oscillatory (ENO) and weighted essentially non oscillatory (WENO) schemes [5] and discontinuous Galerkin (DG) methods [6]. Moreover, Parsani et al. demonstrated the importance of entropy in enforcing consistent boundary conditions and numerical fluxes to improve numerical accuracy and robustness of the solution [7].

The present work focuses on enhancing shock capturing methodologies under physical and numerical near vacuum conditions when entropy thresholding is employed to detect troubled elements. It is well established that shock capturing presents significant challenges in the simulation of hyperbolic and hyperbolic parabolic conservation laws. These problems are characterized by the presence of discontinuities, such as shock waves and contact discontinuities, which can lead to substantial numerical difficulties. Traditional numerical methods often generate non physical oscillations near these discontinuities, compromising the accuracy and stability of the simulation. The Discontinuous Galerkin method, recognized for its high order accuracy and flexibility in handling complex geometries, is particularly susceptible to these issues. Therefore, effective shock capturing techniques are crucial in this framework.

Several strategies have been developed to address the challenges of shock capturing in DG methods. These include filtering [8–10], limiting [11–13], and artificial viscosity methods [14–16]. Each approach has distinct principles and mechanisms for mitigating non physical oscillations while preserving the essential characteristics of the solution. In the context of shock capturing methods, entropy has also been used to stabilize the numerical solution. Indeed, Guermond et al. [17] introduced the concept of entropy viscosity to smear numerical discontinuities. The usage of entropy has been extended to shock sensing in

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previous works [14,18] with discrete success. The current work builds upon previous advancements in modal filtering within the parameter free shock capturing scheme of D'Afiero et al. [19], improving robustness near vacuum conditions with a modified entropy functional that exhibits better numerical behavior in the low isentropic pressure regime.

2. Discontinuous spectral element methods

The governing equations are discretized using a nodal discontinuous spectral element method. The computational domain Ω is decomposed into N_e non overlapping elements Ω_k, such that

$$\Omega = \bigcup_{k=1}^{N_e} \Omega_k. \tag{1}$$

Each physical element is obtained from a reference element $\hat{\Omega}$ through the mapping

$$\mathbf{x} = \mathbf{X}_k(\boldsymbol{\zeta}), \tag{2}$$

where $\boldsymbol{\zeta}$ denotes the reference coordinates. The Jacobian matrix and its determinant are defined as

$$\mathbf{J}_k = \frac{\partial \mathbf{X}_k}{\partial \boldsymbol{\zeta}}, \qquad J_k = \det(\mathbf{J}_k). \tag{3}$$

The metric terms associated with this mapping are included in the evaluation of the flux divergence.

Let Q denote the set of solution points in the reference element. Within each element, the conservative state vector is approximated by the nodal polynomial expansion

$$\mathbf{q}_h^k(\mathbf{x}, t) = \sum_{j \in Q} \mathbf{q}_j^k(t) \phi_j(\boldsymbol{\zeta}), \tag{4}$$

where $\mathbf{q}_j^k$ is the vector of conservative variables at the solution point $\boldsymbol{\zeta}_j$, and ϕ_j are the nodal basis functions. The basis functions satisfy

$$\phi_j(\boldsymbol{\zeta}_i) = \delta_{ij}. \tag{5}$$

The numerical solution is approximated using polynomials of degree p, i.e. $\mathbf{q}_h^k \in \mathbb{P}_p$.

The physical flux divergence is evaluated through its reference space formulation. If $\mathcal{F}_m$ denotes the physical flux in the x_m direction, the contravariant flux in the reference direction ζ_α is defined as

$$\widetilde{\mathcal{F}}_\alpha = J_k \sum_{m=1}^{N_d} \frac{\partial \zeta_\alpha}{\partial x_m} \mathcal{F}_m, \qquad \alpha = 1, \dots, N_d, \tag{6}$$

where N_d is the spatial dimension. Hence,

$$\nabla_{\mathbf{x}} \cdot \mathcal{F} = \frac{1}{J_k} \sum_{\alpha=1}^{N_d} \frac{\partial \widetilde{\mathcal{F}}_\alpha}{\partial \zeta_\alpha}. \tag{7}$$

The reference differentiation matrices are defined as

$$D_{ij}^{(\alpha)} = \frac{\partial \phi_j}{\partial \zeta_\alpha}(\boldsymbol{\zeta}_i), \qquad i, j \in Q, \qquad \alpha = 1, \dots, N_d. \tag{8}$$

The volume contribution to the residual at solution point i of element Ω_k is therefore written as

$$(\nabla \cdot \mathcal{F})_{\Omega_k, i} = \frac{1}{J_i^k} \sum_{\alpha=1}^{N_d} \sum_{j \in Q} D_{ij}^{(\alpha)} \widetilde{\mathcal{F}}_{\alpha, j}^k. \tag{9}$$

Thus, the volume operator used in compact notation corresponds to the differentiation matrices $D_{ij}^{(\alpha)}$, together with the metric terms and the multiplication by $1/J_i^k$.

The coupling between neighboring elements is introduced through numerical fluxes at the element faces. Let S_k be the set of faces of Ω_k, and let I_f denote the set of quadrature points on face $f \in S_k$. The interior and exterior traces at a face point are denoted by $\mathbf{q}_r^{k,-}$ and $\mathbf{q}_r^{k,+}$, respectively. These two traces are evaluated at the same physical face point. If the face quadrature points do not coincide with the solution points, the traces are obtained by interpolation from the solution points to the face quadrature points.

The interpolation matrix from solution points to face quadrature points is defined as

$$V_{rj}^f = \phi_j(\boldsymbol{\zeta}_r^f), \qquad r \in I_f, \qquad j \in Q. \tag{10}$$

The element mass matrix and the face quadrature matrix are

$$M_{ij} = \int_{\hat{\Omega}} \phi_i(\boldsymbol{\zeta}) \phi_j(\boldsymbol{\zeta})\, d\boldsymbol{\zeta}, \qquad W_{rs}^f = \omega_r^f \delta_{rs}, \tag{11}$$

where ω_r^f are the face quadrature weights. The lifting operator that maps face residuals back to the solution points is

$$L^f = M^{-1} \left(V^f\right)^T W^f. \tag{12}$$

The common normal flux at a face point is denoted by

$$\mathcal{F}_{n,r}^{*,f} = \mathcal{F}_n^* \left(\mathbf{q}_r^{k,-}, \mathbf{q}_r^{k,+}, \mathbf{n}_r^{k,f}\right), \tag{13}$$

where $\mathbf{n}_r^{k,f}$ is the outward unit normal vector. The inviscid part of the common flux is computed using an exact or approximate Riemann solver. The surface contribution to the residual is then written as

$$(\nabla \cdot \mathcal{F})_{\partial\Omega_k, i} = \frac{1}{J_i^k} \sum_{f \in S_k} \sum_{r \in I_f} L_{ir}^f \left[\mathcal{F}_{n,r}^{*,f} - \mathcal{F}_{n,r}^{-,f}\right], \tag{14}$$

where $\mathcal{F}_{n,r}^{-,f}$ is the normal flux obtained from the interior polynomial trace. The surface term therefore corrects the discontinuous interior normal flux with the common numerical flux at the interface.

Combining the volume and surface contributions gives the semi discrete system

$$\frac{d\mathbf{q}_i^k}{dt} = -\left[(\nabla \cdot \mathcal{F})_{\Omega_k, i} + (\nabla \cdot \mathcal{F})_{\partial\Omega_k, i}\right]. \tag{15}$$

For Navier Stokes computations, the viscous terms are discretized using a local discontinuous Galerkin formulation. Since the viscous operator is of second order, an auxiliary gradient variable is introduced as

$$\mathbf{G} = \nabla \mathbf{w}, \tag{16}$$

where $\mathbf{w}$ denotes the variables required to evaluate the viscous fluxes. The viscous flux can then be expressed as

$$\mathcal{F}^V = \mathcal{F}^V(\mathbf{q}, \mathbf{G}). \tag{17}$$

The auxiliary gradient equation is discretized using the same volume differentiation and lifting operators introduced above.

At a face point, the LDG common solution is defined as

$$\mathbf{w}^* = \left(\frac{1}{2} - \beta\right) \mathbf{w}^- + \left(\frac{1}{2} + \beta\right) \mathbf{w}^+, \tag{18}$$

where β controls the orientation of the LDG trace. The reconstructed gradient at the solution points is written as

$$\mathbf{G}_i^k = \nabla_h \mathbf{w}_i^k + \sum_{f \in S_k} \sum_{r \in I_f} L_{ir}^f \left(\mathbf{w}_r^{*,f} - \mathbf{w}_r^{-,f}\right) \mathbf{n}_r^{k,f}, \tag{19}$$

where ∇_h is the element local polynomial gradient computed from the differentiation matrices $D_{ij}^{(\alpha)}$. The second term is the LDG correction associated with the common solution at the faces.

The common normal viscous flux is defined as

$$\mathcal{F}_n^{V,*} = \left[\left(\frac{1}{2} + \beta\right) \mathcal{F}^V(\mathbf{q}^-, \mathbf{G}^-) + \left(\frac{1}{2} - \beta\right) \mathcal{F}^V\left(\mathbf{q}^+, \mathbf{G}^+\right)\right] \cdot \mathbf{n} + \tau\left(\mathbf{w}^- - \mathbf{w}^+\right). \tag{20}$$

The parameter τ controls the interface penalty contribution. In the LDG formulation, when the common solution is biased towards one side of the interface, the common viscous flux is biased in the opposite direction. In all viscous computations presented in this work, the LDG parameters are fixed to

$$\beta = 0.5, \qquad \tau = 0.1. \tag{21}$$

The resulting system of ordinary differential equations is advanced in time using an explicit strong stability preserving Runge–Kutta method under a suitable CFL restriction. The modified entropy functional introduced later in this work is not used to define the viscous numerical fluxes, the LDG gradient reconstruction or an entropy stable Navier Stokes discretization. Its role is limited to the troubled element detection procedure used to activate the modal shock capturing filter.

2.1. Modal filtering

The solution approximation written in a nodal basis can alternatively be expressed through a modal expansion [20]. For a generic element Ω_k, the conservative state is written as

$$\mathbf{q}_h^k(\mathbf{x},t) = \sum_{i=0}^{N_p-1} \hat{\mathbf{q}}_{k,i}(t)\psi_i(\mathbf{x}), \tag{22}$$

where $N_p = \dim(\mathbb{P}_p)$ is the number of modal basis functions, $\psi_i(\mathbf{x})$ are modal basis functions, such as Legendre polynomials, and $\hat{\mathbf{q}}_{k,i}$ are the corresponding modal coefficients of the conservative variables. The modal basis is chosen to be orthogonal with respect to the element inner product.

A modal filter is introduced by multiplying each modal coefficient by a scalar damping coefficient. The filtered solution is therefore written as

$$\widetilde{\mathbf{q}}_h^k(\mathbf{x},t) = \sum_{i=0}^{N_p-1} H_{k,i}(t)\hat{\mathbf{q}}_{k,i}(t)\psi_i(\mathbf{x}), \tag{23}$$

with

$$0 \le H_{k,i}(t) \le 1. \tag{24}$$

Here, $H_{k,i}(t) = 1$ corresponds to no filtering of mode i, whereas smaller values introduce increasing modal damping. The coefficient $H_{k,i}(t)$ should not be interpreted as a function of the single modal coefficient $\hat{\mathbf{q}}_{k,i}$ alone. In the present method, the amount of filtering is determined from sensors evaluated at the element level, while the modal index i controls how this dissipation is distributed across the modal spectrum. Higher modes are therefore damped more strongly than lower modes. The sensor based construction of $H_{k,i}(t)$ is described in Section 4.

The modal filtering operation is applied locally, element by element. Let $\mathbf{x} = \mathbf{X}_k(\boldsymbol{\zeta})$ denote the mapping from the reference element $\hat{\Omega}$ to the physical element Ω_k, with Jacobian $J_k(\boldsymbol{\zeta})$. The modal coefficients are obtained using the physical, Jacobian weighted inner product. The element mass matrix is

$$(M_k)_{ij} = \int_{\hat{\Omega}} \psi_i(\boldsymbol{\zeta})\psi_j(\boldsymbol{\zeta})J_k(\boldsymbol{\zeta})\,\mathrm{d}\boldsymbol{\zeta}, \tag{25}$$

and the modal coefficients are computed as

$$\hat{\mathbf{q}}_k = M_k^{-1}\int_{\hat{\Omega}} \mathbf{q}_h^k(\mathbf{X}_k(\boldsymbol{\zeta}),t)\boldsymbol{\psi}(\boldsymbol{\zeta})J_k(\boldsymbol{\zeta})\,\mathrm{d}\boldsymbol{\zeta}. \tag{26}$$

The filtered polynomial is then obtained from Eq. (23) by applying the modal damping coefficients $H_{k,i}(t)$.

The filter is dissipative because each modal coefficient is multiplied by a factor bounded between zero and one. Equivalently,

$$\left\| H_{k,i}(t)\hat{\mathbf{q}}_{k,i} \right\| \le \left\| \hat{\mathbf{q}}_{k,i} \right\|. \tag{27}$$

Conservation is enforced by preserving the physical zeroth moment,

$$\int_{\Omega_k} \widetilde{\mathbf{q}}_h^k\,\mathrm{d}\mathbf{x} = \int_{\Omega_k} \mathbf{q}_h^k\,\mathrm{d}\mathbf{x}. \tag{28}$$

Equivalently, in reference coordinates,

$$\int_{\hat{\Omega}} \widetilde{\mathbf{q}}_h^k(\mathbf{X}_k(\boldsymbol{\zeta}),t)J_k(\boldsymbol{\zeta})\,\mathrm{d}\boldsymbol{\zeta} = \int_{\hat{\Omega}} \mathbf{q}_h^k(\mathbf{X}_k(\boldsymbol{\zeta}),t)J_k(\boldsymbol{\zeta})\,\mathrm{d}\boldsymbol{\zeta}. \tag{29}$$

For affine elements, this is achieved by preserving the constant mode. For non affine or curved elements, the metric terms enter through the same Jacobian weighted mass matrix used in the projection, and conservation refers to preservation of the physical element average rather than only the reference element average.

In many filtering methods, the filter is designed to introduce stronger dissipation on high frequency modes [10,20–22]. These modes are most commonly associated with spurious oscillations near discontinuities. Modal filtering therefore aims to damp unresolved oscillatory content while preserving the lower order content of the solution as much as possible.

3. Governing equations

The Navier Stokes equations can be expressed as a system of conservation laws, with the semi discrete form given by:

$$\partial_t \mathbf{q} + \nabla \cdot \left(\mathcal{F}_I(\mathbf{q}) + \mathcal{F}_V(\mathbf{q})\right) = 0, \tag{30}$$

where $\mathcal{F}_I(\mathbf{q})$ and $\mathcal{F}_V(\mathbf{q})$ represent the Eulerian, or hyperbolic, and viscous, or parabolic, flux components, respectively. These components are defined as

$$\mathcal{F}_I = \begin{bmatrix} \rho\mathbf{v} \\ \rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I} \\ (E+p)\mathbf{v} \end{bmatrix}, \qquad \mathcal{F}_V = \begin{bmatrix} 0 \\ -\mu\left(\nabla\mathbf{v} + \nabla\mathbf{v}^T\right) \\ -\mu\left(\nabla\mathbf{v} + \nabla\mathbf{v}^T\right)\mathbf{v} - \mu\frac{\gamma}{\mathrm{Pr}}\nabla e \end{bmatrix}. \tag{31}$$

Here,

$$\mathbf{q} = [\rho, \rho\mathbf{v}, E]^T \tag{32}$$

is the vector of conserved variables, where ρ denotes the fluid density, $\mathbf{v}$ is the velocity vector, and

$$E = \frac{p}{\gamma-1} + \frac{1}{2}\rho\|\mathbf{v}\|^2 \tag{33}$$

is the total energy per unit volume, with p being the pressure. The specific internal energy is denoted by

$$e = \frac{p}{(\gamma-1)\rho}. \tag{34}$$

The Prandtl number is set to $\mathrm{Pr} = 0.71$, and μ is the dynamic viscosity. The Euler equations are recovered in the case where $\mu = 0$. Lastly, $\mathbf{I}$ is the identity matrix.

3.1. Entropy-flux pair

Given the presence of discontinuities in the solution of Eq. (30), we seek a weak solution that satisfies given physical principles, referred to as an entropy solution. Introduce $\hat{\eta}(\mathbf{q})$ as a convex functional of the conserved variables $\mathbf{q}$. For the compressible Euler equations, we have

$$\hat{\eta}(\mathbf{q}) = -\frac{\rho k(\mathbf{q})}{\gamma-1}, \qquad \Phi(\mathbf{q}) = -\frac{\rho v k(\mathbf{q})}{\gamma-1}, \tag{35}$$

where $\hat{\eta}(\mathbf{q})$ is the entropy functional, $\Phi(\mathbf{q})$ is its associated entropy flux, $k(\mathbf{q})$ is an entropy function and v denotes the velocity component in the one dimensional formulation. Lax [23] showed that the entropic solution satisfies the following inequality:

$$\frac{\partial\hat{\eta}(\mathbf{q})}{\partial t} + \frac{\partial\Phi(\mathbf{q})}{\partial x} \le 0. \tag{36}$$

In smooth flow regions, the equality holds, whereas in the vicinity of discontinuities, such as shocks, the inequality is strict. It is important to note that not all functionals $\hat{\eta}(\mathbf{q})$ are valid entropy flux pairs for the governing equations of interest. Lax [23] proved that the entropy functional $\hat{\eta}(\mathbf{q})$ must not only satisfy the entropy inequality of Eq. (36), but the entropy-flux pair must also satisfy a compatibility condition [23], given by

$$\partial_{\mathbf{q}}\Phi(\mathbf{q}) = \partial_{\mathbf{q}}\hat{\eta}\cdot\partial_{\mathbf{q}}f(\mathbf{q}), \tag{37}$$

where $f(\mathbf{q})$ is the flux vector of the system of conservation laws. In this context, for the analytical treatment, the system under consideration is the compressible Euler equations, which, without loss of generality, has been considered in its one dimensional form. In the case of the

Euler equations, which are strictly hyperbolic, the entropy constraint is mathematically well posed and can be used to detect elements neighboring a discontinuity. Assuming a finite speed of propagation of the solution and under a proper Courant Friedrichs Lewy restriction for the time step δt, the local entropy bound is evaluated over the Voronoi neighbors of the current element, which represent its domain of influence. This is expressed as

$$k(\mathbf{q}(\mathbf{x}_i, t+\delta t)) \geq \min_{\mathbf{x}\in\mathcal{V}_k} k(\mathbf{q}(\mathbf{x}, t)), \tag{38}$$

where $\mathbf{x}_i$ is a point in the discretization space and $\mathcal{V}_k$ denotes the local domain of dependence associated with element Ω_k. Troubled elements are detected not only based on the entropy inequality but also using positivity preserving criteria. In the following, the conservative state at solution point $\mathbf{x}_j$ in element Ω_k is denoted by

$$\mathbf{q}_j^k = \mathbf{q}_h^k(\mathbf{x}_j, t).$$

Therefore, violations of the following set of inequalities are considered:

$$\begin{aligned} &p(\mathbf{q}_j^k) - p_{\text{tol}} \geq 0, \\ &\rho(\mathbf{q}_j^k) - \rho_{\text{tol}} \geq 0, \\ &k(\mathbf{q}_j^k) + k_{\text{tol}} \geq \min_{\mathbf{x}\in\mathcal{V}_k} k(\mathbf{q}_h(\mathbf{x}, t)). \end{aligned} \tag{39}$$

These inequalities are evaluated at each solution point $\mathbf{x}_j$ within the element. The symbol $\mathbf{q}$ is used consistently here because pressure, density and the entropy function are all evaluated from the conservative state vector. For further computational details regarding the implementation of the physical inequalities, the reader is referred to Dzanic et al. [18].

3.2. Modified entropy functional

This study presents a piecewise continuously differentiable entropy function $k(\mathbf{q})$, designed to improve numerical behavior under near vacuum conditions, a key aspect for the efficacy of shock capturing methods based on entropy thresholding. The importance of this enhancement becomes particularly apparent when pressure levels in the solution approach zero, resulting in an isentropic pressure function $\xi = p/\rho^\gamma$ that also approaches zero. Under these circumstances, the thermodynamic entropy $\sigma(\xi(\mathbf{q})) = \log(\xi)$ exhibits singular behavior. Consequently, even minor fluctuations in the isentropic pressure, potentially within the roundoff error limits of hardware precision, can lead to significant variations in the computed entropy. Such sensitivity can compromise the reliability of entropy based detection strategies of troubled elements in shock capturing methodologies. It is important to note that a near zero state of the isentropic pressure function is not exclusively associated with vacuum conditions. Indeed, even an ill behaved Riemann solver can locally produce non physical states characterized by negative pressure or density values, or values approximately zero, which must be considered when calculating entropy. Therefore, achieving more robust behavior near $\xi \approx 0$ is highly desirable to enhance the performance of shock capturing methods. In this study, the piecewise entropy function is defined as follows:

$$k(\xi(\mathbf{q})) = \begin{cases} g(\xi) = -\dfrac{(bd+c)^2}{bd(b\xi+c)} + \left(\ln(d) + \dfrac{bd+c}{bd}\right), & 0 < \xi \leq d, \\ \sigma(\xi) = \ln(\xi), & \xi > d. \end{cases} \tag{40}$$

The function $g(\xi)$ is specifically designed to exhibit a non steep behavior for values of ξ approaching zero while mimicking the derivative of the thermodynamic entropy at the matching point. The matching point is determined by C^1 continuity and is defined by d, which has been set to $d = 1$ to ensure a smooth transition. The constants b and c are selected to achieve the desired numerically friendly behavior, with the current work utilizing $b = 1.5$ and $c = 1.3$. A graphical representation of the entropy function $k(\xi)$ is provided in Fig. 1.

It is essential to acknowledge that not all functions are valid entropy-flux pairs for the governing equations of interest. Indeed, the compatibility condition in Eq. (37) and the entropy inequality of Eq. (36) must hold. In numerical studies, it is well known that the thermodynamic entropy exhibits stiff behavior due to the logarithmic function in its definition. A closely related choice has been used in the subcell limiting literature. In particular, Rueda–Ramírez et al. [24] considered the modified specific entropy, or entropy proxy,

$$\eta_R = e\rho^{1-\gamma}, \tag{41}$$

where e is the specific internal energy. For an ideal gas, this quantity is directly proportional to the isentropic pressure,

$$\eta_R = e\rho^{1-\gamma} = \frac{p}{(\gamma-1)\rho}\rho^{1-\gamma} = \frac{1}{\gamma-1}\frac{p}{\rho^\gamma} = \frac{\xi}{\gamma-1}. \tag{42}$$

Therefore, η_R removes the logarithmic singularity of the thermodynamic entropy $\sigma(\xi) = \log(\xi)$ as $\xi \to 0$. In this sense, it addresses the same near vacuum difficulty targeted in the present work. The difference is that η_R is essentially a linear proxy for $\xi = p/\rho^\gamma$. It is therefore well behaved as $\xi \to 0$, but it does not recover the logarithmic thermodynamic entropy away from the near vacuum regime. By contrast, the present piecewise functional is constructed to coincide with $\log(\xi)$ for $\xi > d$, while replacing only the low ξ branch with a regularized expression. Thus, the proposed functional preserves the classical entropy behavior in regular regions and acts as a regularized entropy indicator only in the low isentropic pressure regime. It can be shown that the isentropic pressure ξ obeys the entropy compatibility condition of Eq. (37), but never obeys the entropy inequality of Eq. (36) for any ξ. In this section, we first demonstrate that the piecewise entropy functional introduced obeys the compatibility condition and then analyze the convexity of the proposed functional.

The subsequent analysis for both the compatibility and the convexity focuses on $\xi \leq 1$, since it is established that the thermodynamic entropy $\sigma(\xi)$ is a valid entropy-flux pair obeying both relationships, and the proposed piecewise functional is C^1 continuous.

For the branch $\xi \leq 1$, the entropy functional is defined as

$$\hat{\eta}(\xi) = -\rho g(\xi)/(\gamma-1), \tag{43}$$

having chosen $g(\xi)$ as an entropy representation, $k(\mathbf{q}) = g(\xi(\mathbf{q}))$.

For this formulation, the term $\partial_{\mathbf{q}}\boldsymbol{\Phi}(\hat{\eta}(\xi(\mathbf{q}))) = \partial_{\mathbf{q}}\boldsymbol{\Phi}(\mathbf{q})$ corresponds to the left hand side of Eq. (37). In the following one dimensional derivation, $m = \rho v$ denotes the momentum. We compute this expression explicitly as

$$\partial_{\mathbf{q}}\boldsymbol{\Phi}(\mathbf{q}) = \begin{bmatrix} \dfrac{2m\rho^\gamma(bd+c)^2\left(\gamma\left(2E\rho-m^2\right)-m^2\right)}{d\left(b(\gamma-1)\left(2E\rho-m^2\right)+2c\rho^{\gamma+1}\right)^2} \\ -\dfrac{\log(d)-\frac{(bd+c)^2}{bd\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)}+\frac{bd+c}{bd}}{\gamma-1} + \dfrac{m^2\rho^{-\gamma}(bd+c)^2}{d\rho\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)^2} \\ -\dfrac{4m\rho^{\gamma+2}(bd+c)^2}{d\left(b(\gamma-1)\left(2E\rho-m^2\right)+2c\rho^{\gamma+1}\right)^2} \end{bmatrix}. \tag{44}$$

The right hand side terms of Eq. (37) have been evaluated individually. Specifically, $\partial_{\mathbf{q}}\hat{\eta}$ is given by

$$\partial_{\mathbf{q}}\hat{\eta} = \begin{bmatrix} -\dfrac{\log(d)-\frac{(bd+c)^2}{bd\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)}+\frac{bd+c}{bd}}{\gamma-1} + \dfrac{\rho(bd+c)^2\left(\frac{b\gamma\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)}{\rho}-\frac{bm^2\rho^{-\gamma}(\gamma-1)}{2\rho^2}\right)}{bd(\gamma-1)\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)^2} \\ \dfrac{m\rho^{-\gamma}(bd+c)^2}{d\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)^2} \\ -\dfrac{\rho\rho^{-\gamma}(bd+c)^2}{d\left(b\rho^{-\gamma}\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)+c\right)^2} \end{bmatrix}^T \tag{45}$$

and $\partial_{\mathbf{q}} f(\mathbf{q})$ by

$$\partial_{\mathbf{q}} f$$

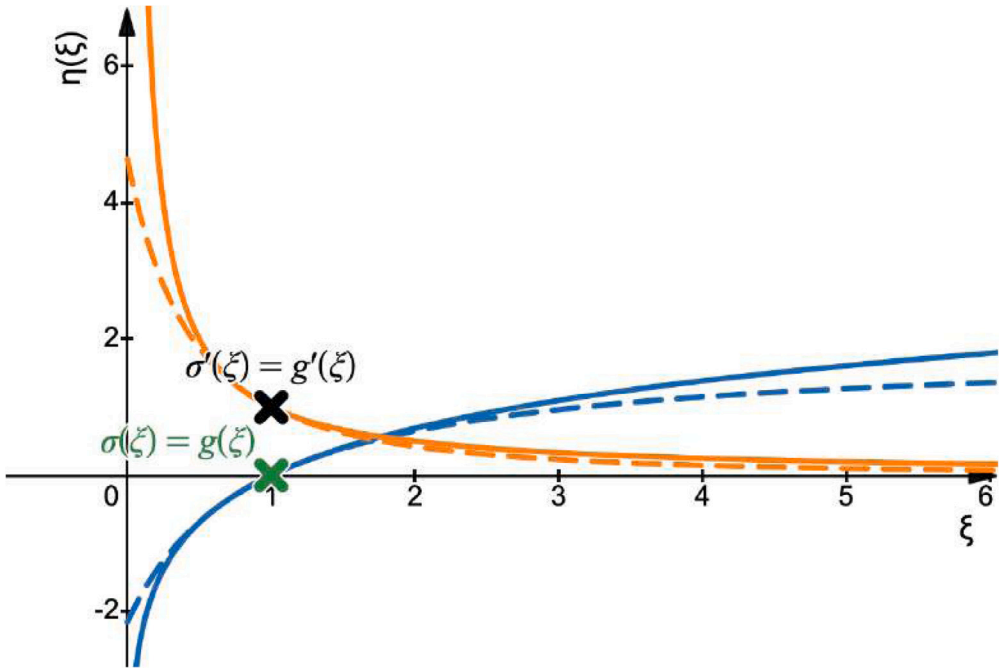


Fig. 1. The classical entropy function $\sigma = \log(p/\rho^\gamma)$ is shown as a solid blue line, while the modified entropy functional $g(\xi)$ is represented by a blue dashed line. The point at which the two functions intersect is indicated by a green cross at $\xi = d$. The derivative of the classical entropy functional, given by $\sigma'(\xi)$, is depicted as a solid orange line, and the derivative of the modified entropy functional, $g'(\xi)$, is shown as an orange dashed line. Their intersection at $\xi = d$ is marked by a black cross.

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{m^2(\gamma-1)}{2\rho^2} - \frac{m^2}{\rho^2} & -\frac{m(\gamma-1)}{\rho} + \frac{2m}{\rho} & \gamma - 1 \\ \frac{m^3(\gamma-1)}{2\rho^3} - \frac{m\left(E+\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)\right)}{\rho^2} & -\frac{m^2(\gamma-1)}{\rho^2} + \frac{E+\left(E-\frac{m^2}{2\rho}\right)(\gamma-1)}{\rho} & \frac{\gamma m}{\rho} \end{bmatrix}. \tag{46}$$

Then we have

$$\partial_{\mathbf{q}} \boldsymbol{\Phi}(\mathbf{q}) = \partial_{\mathbf{q}} \hat{\eta}\, \partial_{\mathbf{q}} f,$$

which completes the proof.

The subsequent analysis focuses on the convexity of the entropy functional. The rationale for seeking a convex functional is formally elaborated in Appendix A, following Lax's proof. Following Harten et al. [1], the convexity conditions can be written directly for the entropy function $g(\sigma)$ as

$$g'(\sigma) > 0, \qquad \frac{g''(\sigma)}{g'(\sigma)} < \frac{1}{\gamma}. \tag{47}$$

The entropy function $g(\xi)$ introduced in this study can be expressed in terms of the thermodynamic entropy $\sigma = \log(\xi)$ as

$$g(\sigma) = -\frac{(bd+c)^2}{bd(be^\sigma + c)} + \left(\ln(d) + \frac{bd+c}{bd}\right), \tag{48}$$

where

$$\xi = e^\sigma. \tag{49}$$

Its first derivative is

$$g' = \frac{(bd+c)^2\, e^\sigma}{d\,(be^\sigma + c)^2}. \tag{50}$$

It is evident that $g' > 0$ provided that $b > 0$ and $d > 0$, which is consistent with the definitions in the current work. The second derivative of $g(\sigma)$ is

$$g'' = \frac{(bd+c)^2\,(-be^\sigma + c)\, e^\sigma}{d\,(be^\sigma + c)^3}. \tag{51}$$

Hence, the ratio of the second derivative to the first derivative gives Harten's condition as

$$\frac{g''}{g'} = \frac{-be^\sigma + c}{be^\sigma + c} < \frac{1}{\gamma}, \tag{52}$$

from which we derive, using Eq. (49), that

$$\xi > \frac{\gamma - 1}{\gamma + 1}\frac{c}{b}. \tag{53}$$

For the specific values of $\gamma = 1.4$, $b = 1.5$, and $c = 1.3$, the lower bound on the convexity of $g(\xi)$ is approximately $\xi \approx 0.1444$. This analysis demonstrates that the introduced functional is not strictly convex for all ranges of ξ. The latter can be achieved if $b \to \infty$, or $\gamma \to 1^+$, which is not of current interest given the ideal gas law assumption, but the numerical stiffness would remain unresolved. Therefore, relaxing the strict convexity requirement is essential for achieving a better behaved numerical entropy. Nevertheless, the aforementioned functional is more firmly grounded in theoretical principles compared to the entropy proxy $\xi = p/\rho^\gamma$, which solely satisfies the compatibility condition but never meets the convexity requirement. The proposed entropy function $g(\xi)$ is a valid entropy flux pair for $\xi \gtrsim 0.144$, but to have a more generalizable behavior over a wider range of test cases, the piecewise definition of the entropy functional $k(\xi)$ in Eq. (40) has been kept. The previous analysis shows that the proposed functional is strictly convex only for $\xi \gtrsim 0.1444$. Below this threshold, the functional no longer satisfies the strict convexity requirement associated with a mathematical entropy in the classical Harten–Lax sense. Consequently, in this low ξ regime, the proposed functional should not be interpreted as providing a global entropy stability proof, nor as defining a rigorous minimum entropy principle. The practical implication is that, when ξ falls below the convexity threshold, the functional is used as a regularized entropy indicator rather than as a strictly convex entropy. This distinction is particularly relevant because the low ξ regime is precisely where the thermodynamic entropy $\log(\xi)$ becomes singular and highly sensitive to small perturbations in pressure or density. The proposed modification therefore trades global convexity for reduced stiffness of the detection variable in near vacuum or strongly under

resolved regions. This loss of convexity does not affect the conservation properties of the underlying DG discretization or of the modal filtering procedure. Moreover, positivity of density and pressure is still enforced independently through the positivity constraints used in the troubled element detection. The role of the modified entropy in the non convex regime is therefore limited to the activation and modulation of the shock capturing/filtering mechanism. In regions where ξ remains above the threshold, the functional retains the interpretation of a valid convex entropy flux pair; below the threshold, it should be regarded as an entropy inspired regularized sensor designed to improve numerical robustness.

4. Shock capturing

4.1. Nonlinear filter design

The nonlinear filtering procedure used in this work is identical to that of D'Afiero et al. [19], including the modal filter, shock sensors and temporal blending of the filter intensity. The only modification introduced here is the replacement of the original thermodynamic entropy used in the troubled element detection criterion with the piecewise entropy functional defined in Eq. (40). For completeness, a brief description of the filter procedure is included.

The filter function used in the modal filtering approach is based on three sensors used to detect discontinuities in the solution: the Ducros sensor f_β [25], the contact discontinuity sensor f_δ [26] and the Persson modal sensor s_e [14]. These sensors are used to automatically tune the filter intensity in flow regions where discontinuities are detected according to the physical admissibility criteria.

The Ducros sensor is defined as

$$f_\beta^{k,n} = \frac{\theta^2}{\theta^2 + \|\boldsymbol{\omega}\|^2 + \varepsilon_s}, \tag{54}$$

where $\theta = \nabla \cdot \mathbf{v}$, $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, and $\varepsilon_s = 10^{-15}$ is a small constant used to avoid division by zero.

The contact discontinuity sensor is given by

$$f_\delta^{k,n} = \frac{\|\nabla\rho\|^2}{\|\nabla\rho\|^2 + \left(\theta^2 + \|\boldsymbol{\omega}\|^2\right)\left(\frac{\rho}{\|\mathbf{v}\|}\right)^2 + \varepsilon_s}. \tag{55}$$

The blending of the Ducros sensor and the contact discontinuity sensor is performed using

$$f_\chi^{k,n} = f_\beta^{k,n} f_\delta^{k,n}. \tag{56}$$

The Persson modal sensor is used to detect unresolved oscillatory content in the element. It is formulated as

$$s_e^{k,n} = \log\left(\frac{\left\langle \chi^{k,n} - \Pi_{p-1}\chi^{k,n}, \chi^{k,n} - \Pi_{p-1}\chi^{k,n}\right\rangle}{\left\langle \chi^{k,n}, \chi^{k,n}\right\rangle + \varepsilon_s}\right), \tag{57}$$

where Π_{p-1} denotes the projection onto the polynomial space of degree $p-1$, and $\chi^{k,n}$ is the scalar quantity used to evaluate the modal smoothness sensor in element Ω_k. In the present computations, $\chi^{k,n} = \rho^{k,n}$.

The threshold value is defined as

$$s_0 = \log\left(1/p^4\right). \tag{58}$$

The smoothness activation function is then

$$f_{s_e}^{k,n} = \frac{1}{1 + \exp\left[-\left(s_e^{k,n} + s_0\right)\right]}. \tag{59}$$

The sensors determine the element level activation of the filter. For a troubled element Ω_k, the instantaneous modal damping coefficient for mode i at Runge–Kutta stage n is defined as

$$h_{k,i}^n = \left(\frac{2\left(1 - f_{s_e}^{k,n}\right)^3}{1 + \exp\left(f_\chi^{k,n}\right)}\right)^{2\deg(\psi_i)}, \tag{60}$$

The quantity $h_{k,i}^n$ is therefore mode dependent: higher-degree modes are damped more strongly through the exponent $2\deg(\psi_i)$.

To avoid abrupt variations of the filter when discontinuities move through the mesh, a temporal blending is applied to each modal coefficient. The difference in filter coefficient between two successive Runge–Kutta stages is defined as

$$\Delta h_{k,i}^n = \left|h_{k,i}^n - h_{k,i}^{n-1}\right|. \tag{61}$$

The blending weights are then

$$\begin{aligned} w_{1,k,i}^n &= 0.5 + 0.5\left(\Delta h_{k,i}^n\right)^{0.25}, \\ w_{2,k,i}^n &= 1 - w_{1,k,i}^n. \end{aligned} \tag{62}$$

The final modal damping coefficient used in Eq. (23) is

$$H_{k,i}^n = w_{1,k,i}^n h_{k,i}^n + w_{2,k,i}^n h_{k,i}^{n-1}. \tag{63}$$

For elements satisfying the physical admissibility criteria in Eq. (39), the modal filter is not activated. This corresponds to setting

$$H_{k,i}^n = 1 \qquad \text{for all modes } i. \tag{64}$$

5. Results

5.1. LeBlanc shock tube

To commence the assessment of the piecewise entropy functional introduced in this study, we examine one of the most challenging numerical regimes defined by the LeBlanc shock tube. This problem is characterized by pressure ratios on the order of $\mathcal{O}(10^9)$ to $\mathcal{O}(10^{10})$ and density ratios on the order of $\mathcal{O}(10^2)$ to $\mathcal{O}(10^3)$, presenting significant challenges for any shock capturing method. The setup includes a right moving shock wave, a left-moving rarefaction wave, and a contact discontinuity. In this work, we consider the LeBlanc problem on the domain $\Omega = [0, 1]$ with Dirichlet boundary conditions. The initial conditions are reported in primitive variables and converted to conservative variables before time integration:

$$(\rho, v, p)(x, 0) = \begin{cases} (1.0, 0.0, p_{\mathrm{L}}), & x < 0.3, \\ (0.01, 0.0, p_{\mathrm{R}}), & x \geq 0.3. \end{cases}$$

The parameters are defined as

$$\gamma = \frac{5}{3}, \qquad p_{\mathrm{L}} = \frac{1}{15}, \qquad p_{\mathrm{R}} = \frac{2}{3 \times 10^{10}}.$$

The solution is advanced up to a final time of $t = 0.5$ time units. The numerical domain has been defined in the range $\Omega \in [0, 1]$ with a varying element count ranging from $N = 50$ to $N = 3200$. To compare the numerical results obtained with the current shock capturing method, a reference solution with an exact Riemann solver has been computed.

To quantify the accuracy of the present numerical solution, we compute the L^1 density error with respect to the exact solution according to

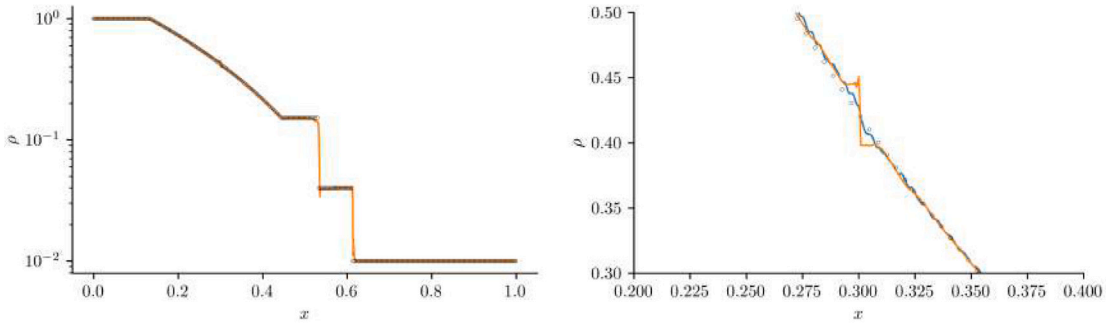
$$\epsilon_\rho^{L^1} = \frac{1}{N_{\mathrm{eval}}} \sum_{i=1}^{N_{\mathrm{eval}}} \left|\rho(x_i) - \rho_{\mathrm{exact}}(x_i)\right|. \tag{65}$$

The convergence behavior was also analyzed across different polynomial degrees, taking advantage of the flexibility of the DG method in enhancing local resolution without remeshing.

This test case exhibits a minimum isentropic pressure based on initial conditions of approximately $\xi \approx 1.44 \times 10^{-7}$ and a maximum of $\xi \approx 0.066$, leading to approximately $\mathcal{O}(10^5)$ orders of magnitude of changes in ξ. This represents an extreme challenge for the entropy thresholding detection used in the current algorithm. In particular, it must also be noted that in this case $\gamma = 5/3$, hence the isentropic

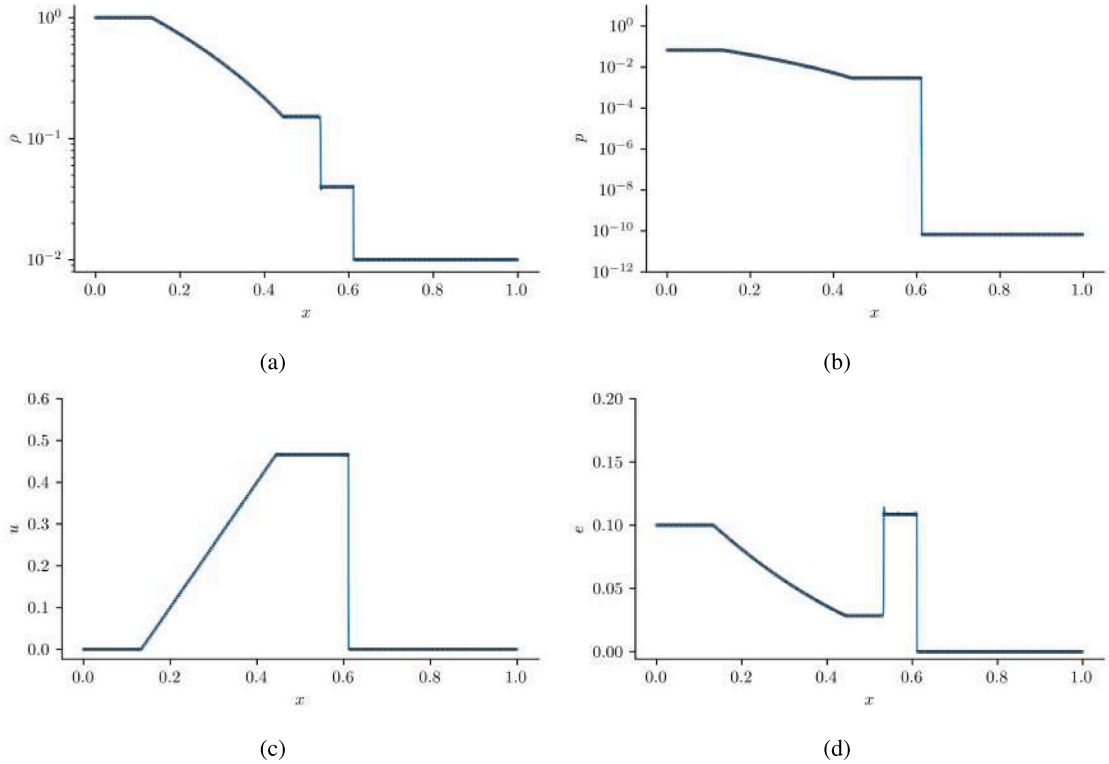
Table 1
Convergence of the L^1 error for the density ϵ_ρ, pressure ϵ_p, velocity ϵ_v, and specific internal energy ϵ_e fields using $p = 3$ and $p = 4$ DG approximations for the LeBlanc shock tube at $t = 0.5$ time units.

N	ϵ_ρ		ϵ_p		ϵ_v		ϵ_e	
	$p = 3$	$p = 4$	$p = 3$	$p = 4$	$p = 3$	$p = 4$	$p = 3$	$p = 4$
50	5.769×10^{-3}	9.503×10^{-3}	4.099×10^{-4}	6.084×10^{-4}	1.376×10^{-2}	1.858×10^{-2}	4.180×10^{-3}	5.437×10^{-3}
100	4.247×10^{-3}	5.110×10^{-3}	2.516×10^{-4}	2.948×10^{-4}	5.854×10^{-3}	1.065×10^{-2}	1.741×10^{-3}	3.266×10^{-3}
200	2.208×10^{-3}	3.344×10^{-3}	1.322×10^{-4}	1.943×10^{-4}	4.537×10^{-3}	6.639×10^{-3}	1.438×10^{-3}	2.077×10^{-3}
400	1.455×10^{-3}	1.638×10^{-3}	8.535×10^{-5}	9.204×10^{-5}	2.540×10^{-3}	3.197×10^{-3}	7.979×10^{-4}	1.004×10^{-3}
800	6.620×10^{-4}	8.622×10^{-4}	3.766×10^{-5}	4.614×10^{-5}	1.239×10^{-3}	1.823×10^{-3}	3.920×10^{-4}	5.731×10^{-4}
1600	3.780×10^{-4}	4.546×10^{-4}	2.153×10^{-5}	2.447×10^{-5}	7.739×10^{-4}	9.211×10^{-4}	2.479×10^{-4}	2.891×10^{-4}
3200	1.989×10^{-4}	2.394×10^{-4}	1.097×10^{-5}	1.256×10^{-5}	3.803×10^{-4}	4.862×10^{-4}	1.287×10^{-4}	1.563×10^{-4}



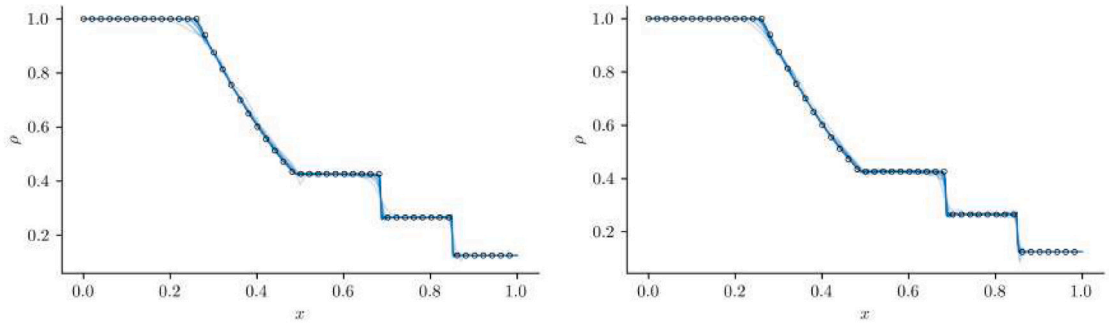
(a) (b)

Fig. 2. LeBlanc shock tube solution at $t = 0.5$ time units using $p = 4$ with $N = 400$ elements. The plots show the numerical solution for density field in Fig. 2(a) and its zoom-in in the region of the initial imposed discontinuity in Fig. 2(b). Blue line, new adaptive entropy functional; orange line, standard entropy functional. Black dots, reference solution.



(a) (b) (c) (d)

Fig. 3. LeBlanc shock tube solution at $t = 0.5$ time units using $p = 4$ and $N = 3200$ elements. The plots show the numerical solution for density Fig. 3(a), pressure Fig. 3(b), velocity Fig. 3(c) and specific internal energy Fig. 3(d). Black dots reference solution.



(a) (b)

Fig. 4. Line plots of the Sod shock tube solution at $t = 0.2$ using $p = 3$ and $p = 4$ discretizations. Each subplot shows the evolution of the solution with increasing mesh resolution, from $N = 25$ to $N = 400$, indicated by increasing opacity. The plots display density, pressure, and velocity profiles, capturing the development of the left-moving rarefaction, contact discontinuity, and right-moving shock. See Fig. 4(a) for $p = 3$ and Fig. 4(b) for $p = 4$ DG approximations.

Table 2
Convergence of the L^1 density error for varying elements N for $p = 2$ to $p = 7$ DG approximations for the Sod shock tube at $t = 0.2$ time units.

N	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$	$p = 7$
25	1.694×10^{-2}	1.294×10^{-2}	1.181×10^{-2}	1.212×10^{-2}	1.023×10^{-2}	1.163×10^{-2}
50	9.575×10^{-3}	7.272×10^{-3}	6.185×10^{-3}	8.004×10^{-3}	5.758×10^{-3}	5.664×10^{-3}
100	4.947×10^{-3}	4.030×10^{-3}	3.567×10^{-3}	4.420×10^{-3}	2.524×10^{-3}	3.347×10^{-3}
200	2.609×10^{-3}	2.063×10^{-3}	1.882×10^{-3}	2.280×10^{-3}	1.539×10^{-3}	1.723×10^{-3}
400	1.400×10^{-3}	1.164×10^{-3}	9.916×10^{-4}	1.266×10^{-3}	9.640×10^{-4}	1.050×10^{-3}

pressure threshold becomes $\xi \approx 0.217$, therefore this test mainly assesses the regularized indicator behavior.

We first consider a comparison with the old entropy functional and the current one for an example resolution of $N = 400$ elements. Notably, in Fig. 2 and its zoomed view in Fig. 2(b), it can be seen that the old entropy functional completely fails in correctly capturing and smoothing the initial discontinuity, which persists in the solution, while the new entropy functional exhibits better behavior. This observation has been consistent across different resolutions for the old entropy functional and, as pointed out earlier, is due to the large changes in ξ near vacuum conditions and throughout the entire domain.

The convergence analysis across varying element counts N and polynomial degree p is shown in Table 1. Convergence is observed across all variables of interest, including pressure, density, velocity, and internal energy. A more representative result, showcasing an oscillation-free solution, is depicted in Fig. 3 for the aforementioned variables. Moreover, it is noteworthy that even in the case of lower resolutions, as in Fig. 2, the shock location was never overpredicted, unlike in previous works [18,27]. Indeed, in those cases, there was a sensible overshoot of the specific internal energy due to a lower estimate of the density, which, in turn, resulted in an overestimate of the shock speed.

5.2. Sod shock tube

The Sod shock tube scenario [28] serves as a benchmark similar to the previous case, with the aim of assessing the scheme's ability to accurately represent shock waves, rarefaction waves, and contact discontinuities. A distinctive aspect of the current case compared to the LeBlanc shock tube is that the isentropic pressure ξ only features changes on the order of $\mathcal{O}(1)$, presenting a different scenario for detection.

The computational domain is defined as $\Omega = [0, 1]$, with the initial state reported in primitive variables:

$$(\rho, v, p)(x, 0) = \begin{cases} (1, 0, 1), & x \leq 0.5, \\ (0.125, 0, 0.1), & \text{otherwise.} \end{cases}$$

Simulations were run until a time of $t = 0.2$ units, and density errors with respect to the exact solution were compared according to Eq. (65). Fig. 4 shows the results obtained with the DG approach with polynomial degrees $p = 3$ and $p = 4$ and varying element counts from $N = 25$ to $N = 400$, with the discretization steps defined according to Table 2. It can be observed that the shock location and the contact discontinuity are predicted consistently despite the resolution in both cases, and a sharp density profile is achieved at the highest $N = 400$ without density oscillations.

The robustness of the underlying method was already proven in a previous work [19] and is confirmed with the current methodology, which, in turn, also features higher accuracy, reaching a minimum error at $N = 400$ and polynomial degree $p = 6$ of $\varepsilon_\rho = 9.64 \times 10^{-4}$, while previously the minimum attained value was $\varepsilon_\rho = 2.49 \times 10^{-3}$ with $N = 400$ and polynomial degree $p = 3$. Therefore, the current methodology not only lowers the error globally, indeed, at $N = 400$ and polynomial degree $p = 3$, we have $\varepsilon_\rho = 1.16 \times 10^{-3}$, but also aids higher order discretization in achieving better accuracy. The overall convergence across varying polynomial degrees and element counts is shown in Table 2.

In the following, the robustness of the current approach is assessed over a variety of standard and challenging test cases, showing that the detection methodology can be improved through the use of the piecewise entropy functional.

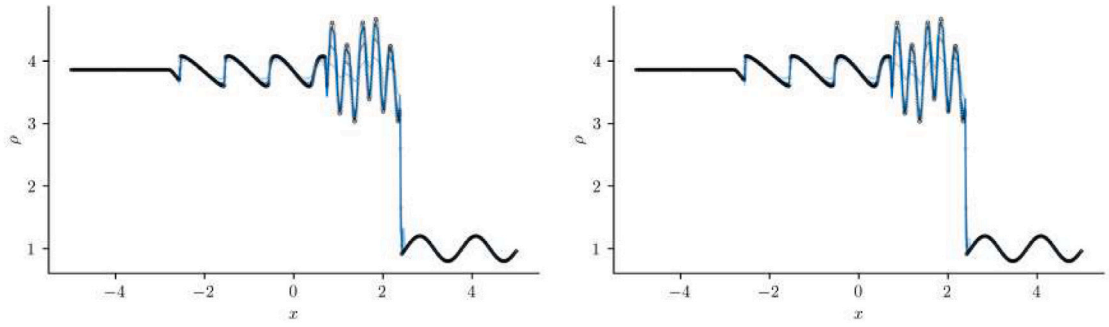
5.3. Shu–Osher problem

To further evaluate the effectiveness of the modified entropy functional in complex scenarios, we examine the Shu and Osher case [29]. This benchmark involves the interaction between a shock wave and an entropy wave, featuring only density changes convected with the flow velocity. The initial condition includes a Mach $M = 3$ shock wave moving into a density field with sinusoidal perturbations. This setup allows for the study of a simplified one dimensional shock turbulence interaction. The problem is defined on the domain $\Omega = [-5, 5]$, with the initial state reported in primitive variables:

$$(\rho, v, p)(x, 0) = \begin{cases} (3.857143, 2.629369, 10.333333), & x \leq -4, \\ (1 + 0.2\sin(5x), 0, 1), & \text{otherwise.} \end{cases}$$

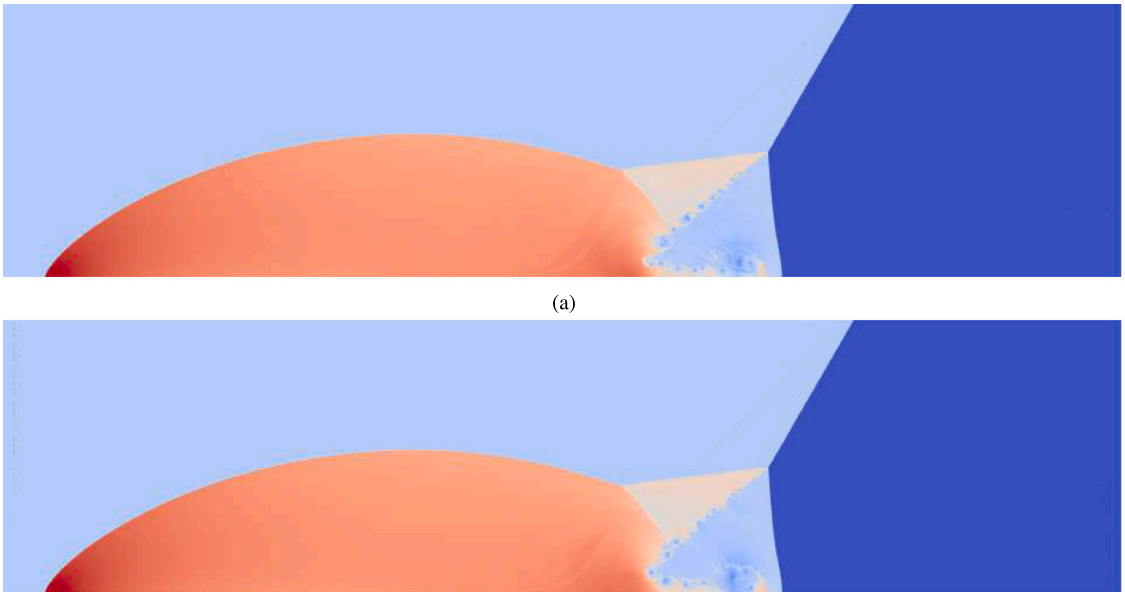
The solutions are compared at a time of $t = 1.8$ units. Instabilities in the solution may emerge in regions of nonlinear interactions, and excessive filtering is often characterized by a poor resolution of the entropy wave, as discussed in [30,31].

Fig. 5 displays the results of the resolved density field using polynomial degrees $p = 3$ and $p = 4$ and varying element counts. The solution is obtained without spurious spikes, while preserving the



(a) (b)

Fig. 5. Line plots of the Shu–Osher problem at $t = 1.8$ using $p = 3$ and $p = 4$ discretizations. Each subplot shows the solution evolution with increasing mesh resolution, with $N = 100$, $N = 200$ and $N = 400$, indicated by increasing opacity. See Fig. 5(a) for $p = 3$ and Fig. 5(b) for $p = 4$ DG approximations.



(a)

(b)

Fig. 6. Density contours in the range $\rho \in [1, 20]$ for the double Mach reflection test case at $t = 0.2$ time units using a $p = 3$ discretization in Fig. 6(a) and $p = 4$ in Fig. 6(b) on a grid consisting of 2400 × 600 elements.

density oscillations behind the shock, thereby improving the baseline results [19].

5.4. Double Mach reflection

The double Mach reflection problem, as introduced by Woodward and Colella [32], serves as an additional test for the proposed filter in scenarios involving strong oblique shocks coexisting with contact discontinuities. This case features a Mach 10 shock impacting a wedge with an angle of $\phi = \pi/6$ at $t = 0$. The numerical domain is defined on a rectangular grid with dimensions $[0, 4] \times [0, 1]$. The grid resolution is 2400 × 600 elements, and polynomial degrees $p = 3$ and $p = 4$ are employed for the discretization. The initial conditions are reported in primitive variables:

$$(\rho, v_x, v_y, p)(\mathbf{x}, 0) = \begin{cases} \mathbf{w}_l, & \text{if } x < \frac{1}{6} + \tan\left(\frac{\pi}{6}\right) y, \\ \mathbf{w}_r, & \text{otherwise,} \end{cases}$$

where

$$\mathbf{w}_l = [8, 7.14471, -4.125, 116.5]^T, \qquad \mathbf{w}_r = [1.4, 0, 0, 1]^T.$$

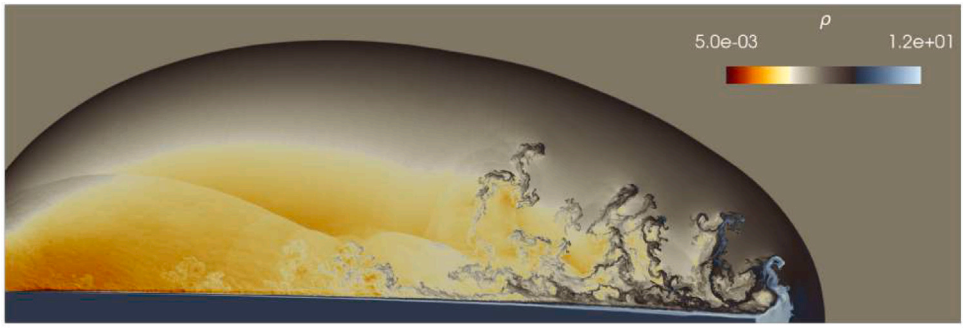
Slip adiabatic wall boundary conditions are applied to the bottom boundary for $x \geq 1/6$. The left boundary and the bottom boundary for $x < 1/6$ enforce the $\mathbf{w}_l$ state. At the right boundary, the solution is set to $\mathbf{w}_r$. For the top boundary, the exact solution is prescribed and is given by

$$(\rho, v_x, v_y, p)(\mathbf{x}, t)_{\text{top}} = \begin{cases} \mathbf{w}_l, & \text{if } x \leq \frac{1}{6} + \tan\left(\frac{\pi}{6}\right) y + \frac{10}{\cos(\pi/6)} t, \\ \mathbf{w}_r, & \text{otherwise.} \end{cases}$$

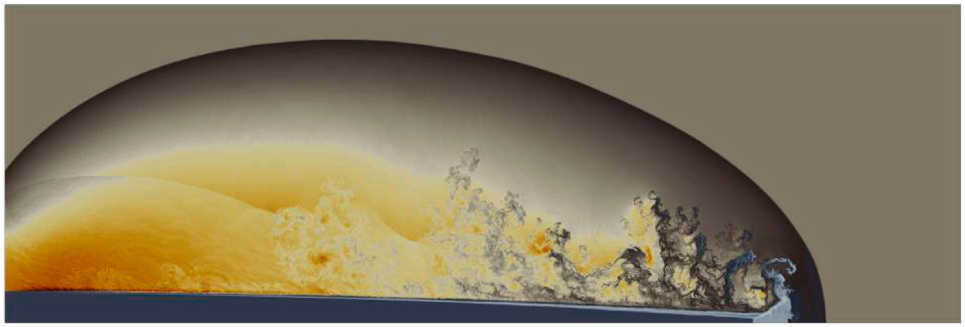
Fig. 6 shows the results for the current approach with polynomial degrees $p = 3$ and $p = 4$. The results are in agreement with the previous work [19], and a slightly finer resolution of smaller scale structures is present at the contact discontinuity. This further confirms the effectiveness of the current methodology.

5.5. Astrophysical jet

The astrophysical jet scenario is employed as an extreme test case to assess the satisfaction of the physical constraints delineated in Eq. (39) and to push the Riemann solver to its performance limits, potentially resulting in unphysical states that challenge the detection algorithm.



(a)



(b)



(c)

Fig. 7. Density contour for the astrophysical jet on a 2400 × 800 grid with $p = 4$ DG approximation at an inflow Mach number of $M = 800$ in Fig. 7(a), $M = 2000$ in Fig. 7(b) and $M = 10^4$ in Fig. 7(c). The flowfield has been rotated such that the positive y axis extends from left to right.

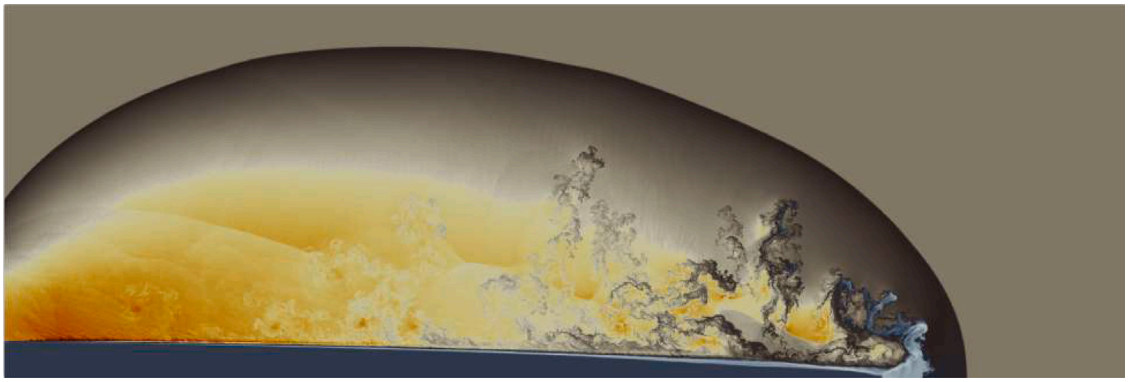
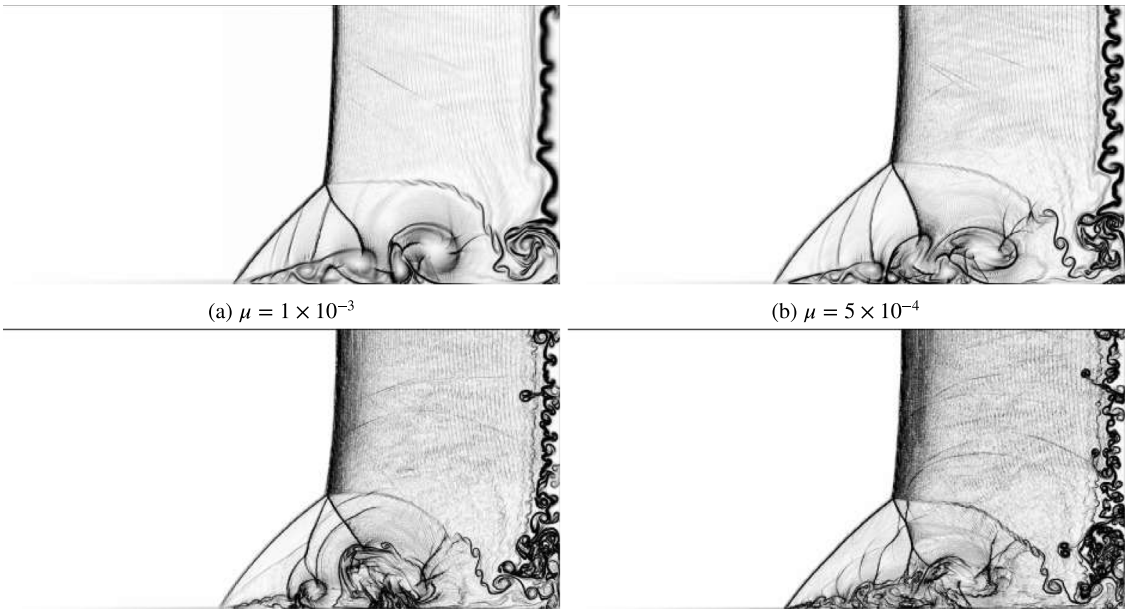


Fig. 8. Density contour for the astrophysical jet on a 2400 × 800 grid with $p = 6$ DG approximation at an inflow Mach number of $M = 10^4$. The flowfield has been rotated such that the positive y axis extends from left to right. The colorbar is the same as in Fig. 7.



(a) $\mu = 1 \times 10^{-3}$ (b) $\mu = 5 \times 10^{-4}$

(c) $\mu = 2 \times 10^{-4}$ (d) $\mu = 1 \times 10^{-4}$

Fig. 9. Schlieren contour for the viscous shock tube problem discretized on a grid consisting of 800 × 400 elements approximated with $p = 4$ polynomials for different dynamic viscosities.

This particular case was initially introduced by Balsara [33]. The setup for this study is adapted from the work of Wu and Shu [34], with the magnetic field being neglected [18,24,35]. The domain under consideration is a half domain $\Omega = [0, 0.5] \times [0, 1.5]$, with symmetry boundary conditions applied along the y axis. The initial conditions within the domain are reported in primitive variables as

$$(\rho, v_x, v_y, p) = [0.1\gamma, 0, 0, 1]^T.$$

For the boundary at $y = 0$, an inlet region is defined for $x \leq 0.05$ with the primitive state

$$(\rho, v_x, v_y, p) = [\gamma, 0, V, 1]^T,$$

yielding a Mach number of $M = V$ relative to the inflow gas. Dirichlet boundary conditions are imposed on the remaining boundary regions. The grid resolution for all the cases analyzed is set to 2400 × 800 elements and is unlikely sufficient to resolve all the scales of the problem. The purpose of this benchmark is therefore to test the capability of the numerical setup to achieve stable and carbuncle free solutions.

This study explores three different Mach numbers: $M = 800$, $M = 2000$, and $M = 10^4$. The solutions for these cases are advanced in time up to final times of $t = 0.002$, $t = 0.0008$, and $t = 0.00016$ time units, respectively. The analysis is first conducted using polynomial degree $p = 4$ to evaluate the robustness of the method at these extreme inflow Mach numbers. Previous research [19] has already established the robustness of the case at $p = 3$ for the $M = 800$ case. The results for the three Mach numbers are shown in Fig. 7. The computation remains stable and fine scale structures are visible on the chosen grid, although no claim of full scale resolution is made.

Utilizing the current piecewise entropy functional, from the analysis of the simpler Sod shock tube, it was observed that the performance of higher polynomial degrees is now comparable to that of the lower order polynomials used previously. Therefore, to confirm the improvements of the proposed approach in scenarios characterized by even lower numerical diffusion of the underlying DG scheme, polynomial degree $p = 6$ was tested in the most demanding case proposed for the current benchmark: the $M = 10^4$ inflow condition. The results are shown in Fig. 8. This outcome is promising for achieving stable, very high order simulations, even in the most challenging contexts, of which a representative sample is presented in the current study.

5.6. Viscous shock tube

The viscous shock tube, introduced by Daru and Tenaud [36], is employed to assess the capability of the proposed methodology in managing intricate shock boundary layer interactions, which are crucial for practical applications. The computational domain spans $\Omega = [0, 1] \times [0, 0.5]$, with the initial condition reported in primitive variables:

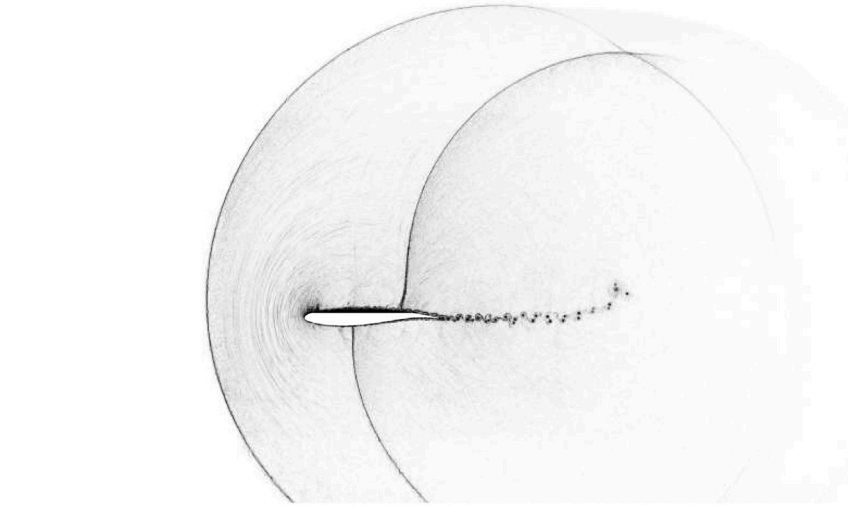
$$(\rho, v_x, v_y, p)(\mathbf{x}, 0) = \begin{cases} (120, 0, 0, 120/\gamma), & x \leq 0.5, \\ (1.2, 0, 0, 1.2/\gamma), & \text{otherwise.} \end{cases}$$

A symmetry boundary condition is applied to the top boundary, while no-slip adiabatic wall conditions are enforced on the remaining boundaries. This scenario encompasses the primary characteristics of classical Riemann problems, including robust and transient shocks, contact discontinuities, and rarefaction waves. This interaction produces a complex lambda shock impacting a viscous boundary layer, thereby offering a suitable test for shock boundary layer interactions.

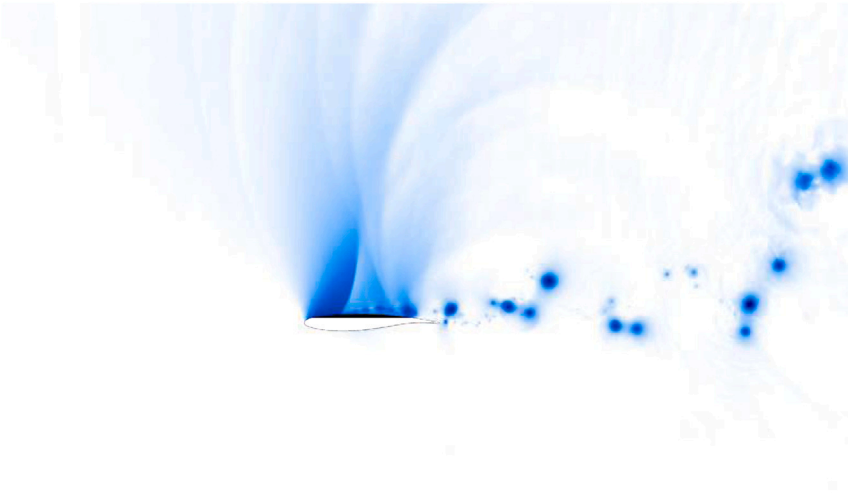
Numerical Schlieren fields at $t = 1$ unit, computed using a DG approximation of polynomial degree $p = 4$ on a uniform 800 × 400 grid, are illustrated in Fig. 9 for different values of the dynamic viscosity μ. As the Reynolds number increases, progressively finer structures emerge in the solution, especially along the contact discontinuity, which are usually filtered out in other, more dissipative, approaches [18,37–39]. Both the lambda shock and the small vortical structures are well resolved and predicted results show good agreement with the previous findings based on entropy filtering [19]. These observations confirm that the proposed methodology preserves its robustness.

5.7. Transonic OAT15a

To assess the effectiveness of the current methodology, particularly when applied to fully unstructured grids, an analysis was conducted on an under resolved OAT15a supercritical airfoil. The conditions for this analysis were characterized by a nominal Mach number of $M =$



(a)



(b)

Fig. 10. Flowfields for the transonic OAT15a supercritical airfoil at $M = 0.73$, $Re = 3 \times 10^6$ and $\alpha = 7°$ approximated with $p = 6$ polynomials. Fig. 10(a) depicts the numerical schlieren at a nondimensional time of $FTT = 1.5$ and 10(b) depicts the density field at a nondimensional time of $FTT = 20$.

0.73, an angle of attack of $\alpha = 7°$, and a chord Reynolds number of $Re = 3 \times 10^6$. This specific case has been extensively examined in previous research, both numerically and experimentally, as detailed in several studies [40–44]. The computational domain for this analysis consisted of a two dimensional C type mesh grid, extending from 6 chord lengths upstream of the airfoil's leading edge to 8 chord lengths downstream of the trailing edge. This grid configuration was previously utilized in another study [19] employing polynomial degree $p = 4$. The primary aim of this investigation was to evaluate the ability of the current methodology to capture the inherently unsteady behavior of the suction side shock using higher polynomial degrees, given the proposed improvements. Indeed, the purpose of the test is to assess the robustness of the proposed filtering strategy on a high Reynolds number transonic configuration involving an unstructured grid, curved geometry, shock motion, and shock induced vortical structures.

The simulation is advanced in time from a cold startup with a constant flow field to achieve the desired Mach and Reynolds numbers. The initial transient at a flow time unit $FTT = 1.5$ is illustrated using numerical schlieren in Fig. 10(a), where a detailed resolution of flow features can be observed. This resolution is further confirmed in the subsequent development at a later time of $FTT = 20$ in Fig. 10(b), which represents the density field. The higher coherence of the shedded vortical structures is characteristic of two dimensional turbulent flows induced by the suction side shock and has been noted in previous two dimensional subsonic and transonic studies [45–47].

6. Conclusion

This study presented a new piecewise entropy functional designed to improve entropy based troubled element detection in high order shock capturing methods, particularly in demanding near vacuum regimes. Building on the generalized entropy framework of Harten, the proposed functional was shown to behave as a convex entropy-flux pair above a computable threshold. Below this threshold, where strict convexity is lost, the functional is used as a regularized entropy indicator rather than as the basis for a global entropy stability proof. A comprehensive theoretical analysis was carried out to examine the mathematical properties of the proposed continuously differentiable entropy functional, with emphasis on convexity and compatibility. The analysis revealed that the new functional is more solidly rooted in theoretical principles than the conventional isentropic pressure. One of the primary advantages of this formulation is its capacity to improve the identification of troubled elements in shock capturing schemes. This improvement is particularly relevant in the context of Discontinuous Galerkin methods, where recent developments have led to parameter

free shock capturing approaches based on the detection of entropy inequality violations. A central objective of this work was to provide a robust mechanism for addressing both near vacuum conditions and poor behaviors stemming from ill conditioned Riemann solvers in extreme flow scenarios. By reinforcing the reliability of troubled element detection, the proposed entropy functional contributes to more accurate and stable numerical computations. The effectiveness of the proposed approach was assessed using a range of representative test cases involving both inviscid and viscous flows. In particular, the limitations of the classical thermodynamical entropy in reproducing the dynamics of the LeBlanc shock tube, a canonical and highly challenging near vacuum benchmark, were shown. Additionally, the capability of the method in hyperbolic contexts was validated through the stable simulation of an astrophysical jet at a Mach number $M = 10^4$, using sixth order polynomial approximations. Its relevance in hyperbolic parabolic regimes was also demonstrated by a transonic flow over an airfoil simulated at polynomial order $\mathbb{P}_6$. Ongoing work is directed towards improving the behavior of the present methodology for steady bow shock configurations, for which the current entropy based filtering strategy still requires further refinement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Convexity of the functional

The convexity requirement for the entropy function, denoted as $\hat{\eta}(u)$, within the framework of the Euler equations, is used for selecting physically admissible weak solutions. In nonlinear systems, such as those aforementioned, initially smooth solutions can develop discontinuities known as shocks, leading to the breakdown of classical solutions. This necessitates the introduction of weak solutions, which extend the concept of a solution to include these singularities. However, since weak solutions are generally not unique, additional admissibility criteria are required. The convexity of the entropy function ensures that, among all possible weak solutions, only those satisfying the entropy inequality given by Eq. (36) are considered physically meaningful. This inequality encodes the second law of thermodynamics, which demands that entropy should increase across shocks, reflecting the irreversible production of entropy.

Following Lax, one considers the viscous regularization of the Euler equations given by

$$\partial_t u + \partial_x f(u) = \varepsilon \partial_{xx} u, \tag{66}$$

where the small viscosity parameter ε tends to zero. The solutions to this regularized problem converge to weak solutions of the original system only if the entropy function $\hat{\eta}(u)$ is convex. This convexity is needed in establishing the entropy inequality

$$\partial_t \hat{\eta}(u) + \partial_x F(u) \leq 0, \tag{67}$$

which emerges as the limit of

$$\partial_t \hat{\eta}(u) + \partial_x F(u) = \varepsilon \cdot \Pi, \tag{68}$$

where the term Π is nonnegative and arises due to the positive definiteness of the Hessian matrix $D^2_{xx}\hat{\eta}(u)$, thus reflecting the strict convexity of $\hat{\eta}$.

Appendix B. Compatibility condition

Consider a conservation law given by

$$\partial_t u + \partial_x f(u) = 0. \tag{69}$$

Define $\hat{\eta}(u)$ be a smooth scalar mathematical entropy function. Differentiating $\hat{\eta}(u)$ with respect to time gives

$$\partial_t \hat{\eta}(u) = \partial_u \hat{\eta}(u)\, \partial_t u. \tag{70}$$

Substituting Eq. (69) into the above yields

$$\partial_t \hat{\eta}(u) + \partial_u \hat{\eta}(u)\, \partial_x f(u) = 0. \tag{71}$$

Assuming that $\hat{\eta}(u)$ satisfies a conservation law, there must exist an associated entropy flux $\Phi(u)$ such that

$$\partial_t \hat{\eta}(u) + \partial_x \Phi(u) = 0. \tag{72}$$

By applying the chain rule to both $\partial_x f(u)$ and $\partial_x \Phi(u)$, the consistency of the two forms of the entropy evolution equations implies the following compatibility condition:

$$\partial_u \Phi(u) = \partial_u \hat{\eta}(u)\, \partial_u f(u). \tag{73}$$

Data availability

Data will be made available on request.

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